A0/A1 fine-aim sensor mount: 3D views

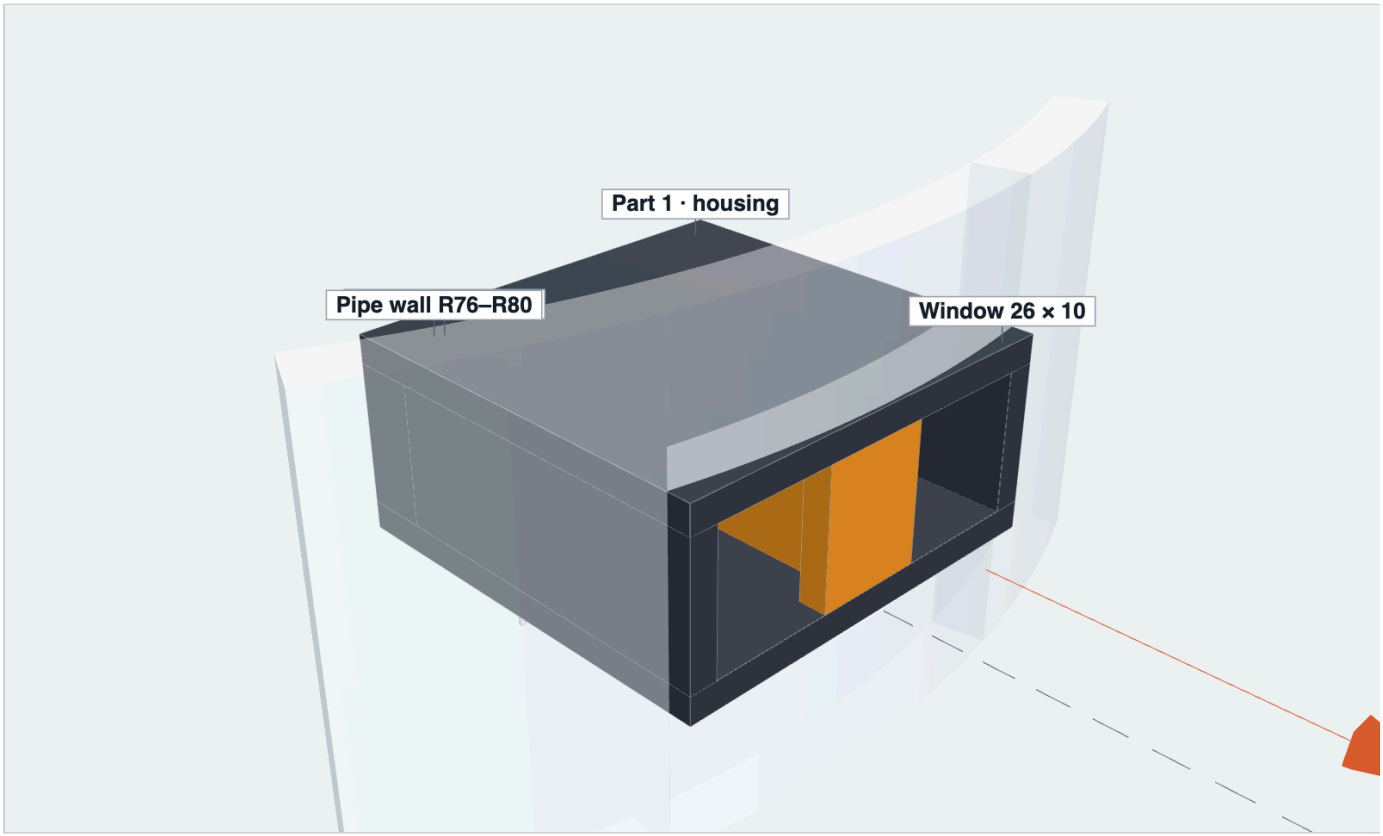
MQ Sentinel · ENGG2000 V3 · Drawing MQS-V3-SNS-001 Rev A (draft) · 2026-10-25 · Prepared by Y. Cui · units mm

Draft for design review. Geometry is calculated from the TSSP58P38 datasheet and Appendix 1 of the brief; it has not been bench-tested. The model draws the front face flat for clarity; the printed part follows the R80 pipe curve.

Colour key: Part 1 housing / Part 3 cover (matte black PLA) Part 2 T-insert TSSP58P38 body Lens PVC pipe wall. Dashed line = laser direction (Datum A); orange arrow = toward the target.

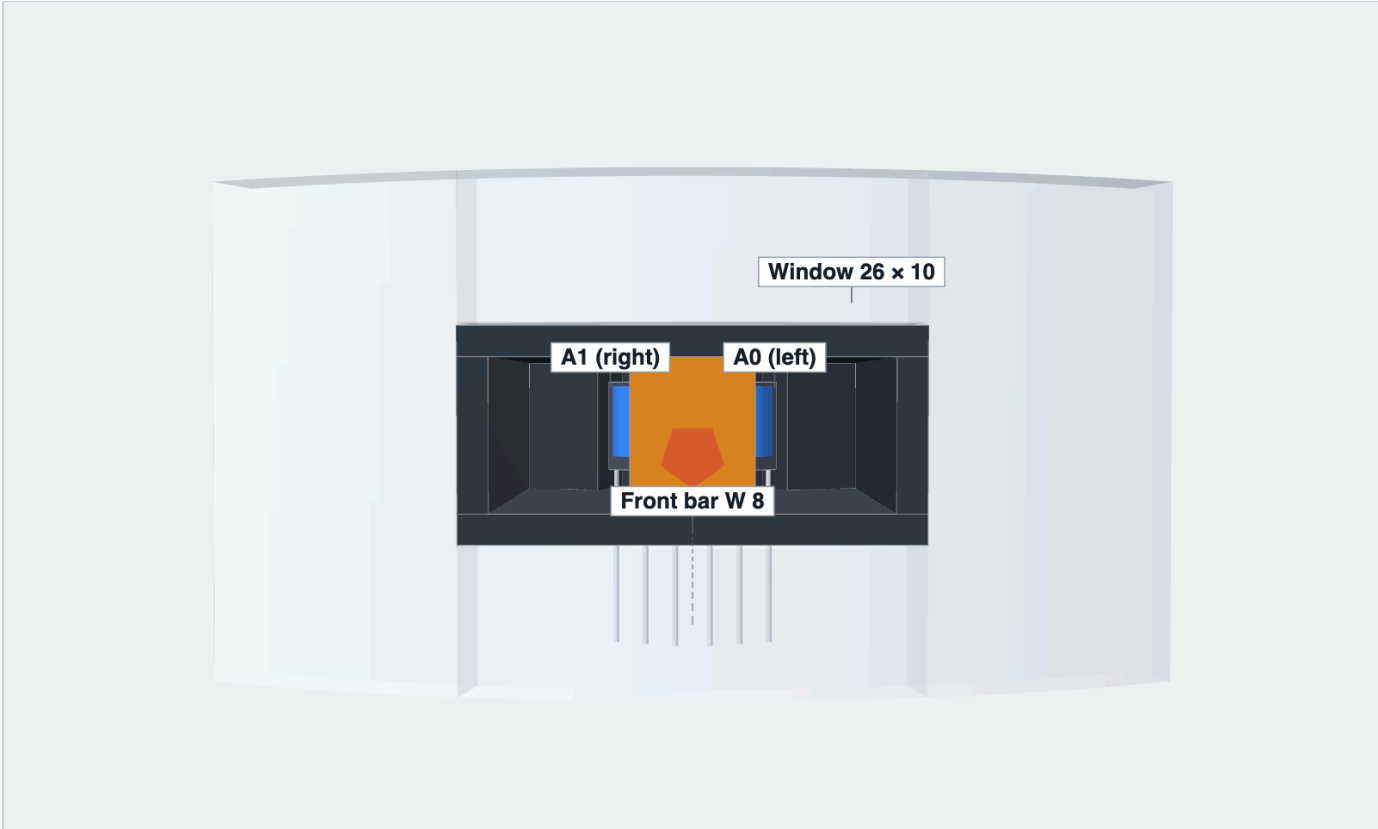
Part	What it does	Size
1 Housing	Holds A0/A1 turned 20° outward, 10 mm apart. Front frame = 26 × 10 window, flush with the pipe outside.	30 × 14 × 29.5
2 T-insert	Stem blocks light crossing the centre; the 8 mm front bar shades the inner edge of each lens (≈0.9 mm), so a small turn changes L and R in opposite directions. Swappable: W 7 / 8 / 9.	stem 3 × 10 × 19, bar W × 10 × 2
3 Top cover	Closes the top and holds Part 2 in its slot; 2 × M2 into the side walls.	30 × 2 × 29.5
A0 / A1	TSSP58P38 receivers, left / right. Leads exit downward through the floor.	5.0 × 6.95 × 4.8

1. Assembled, 3/4 view



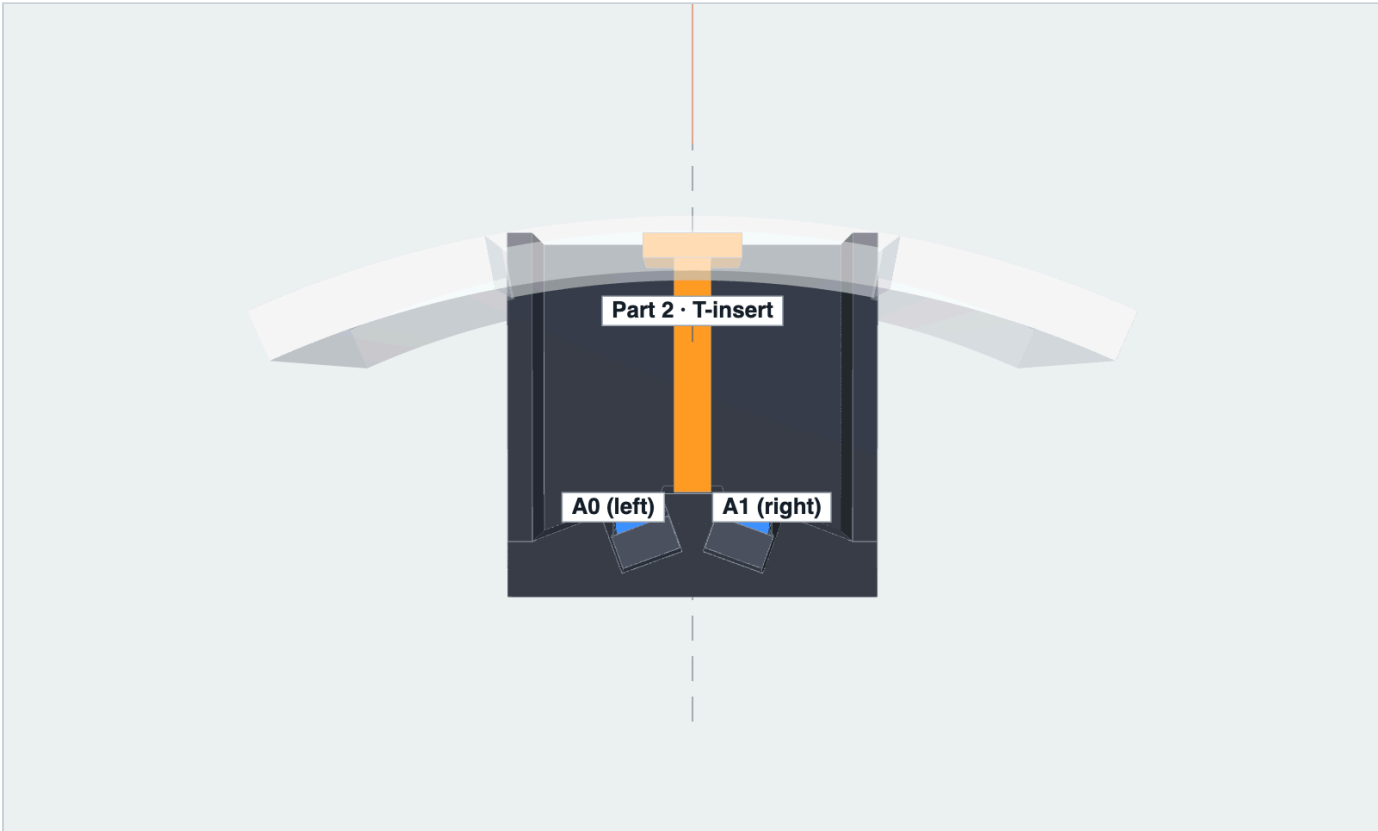
The complete mount sitting in its window in the pipe wall. The front of the housing and the bar are flush with the pipe outside; everything else is inside the pipe.

2. From the target (what the beacon sees)



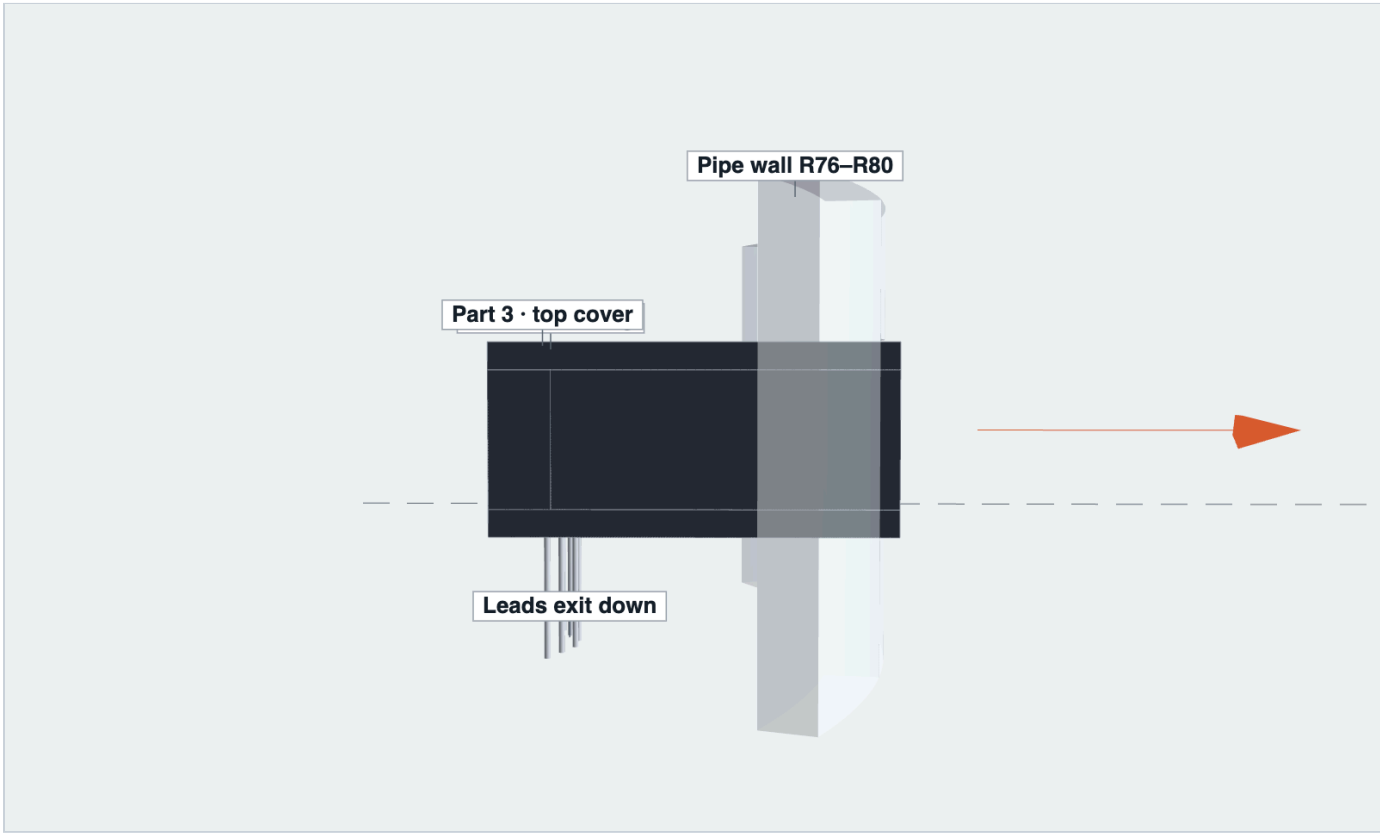
Looking back at the satellite from the target. The 26 × 10 window shows both lenses; the orange bar covers a thin strip (≈0.9 mm) of the inner edge of each lens. Seen from here, A1 (right sensor) is on the viewer's left.

3. Top view, cover removed (target at the top)



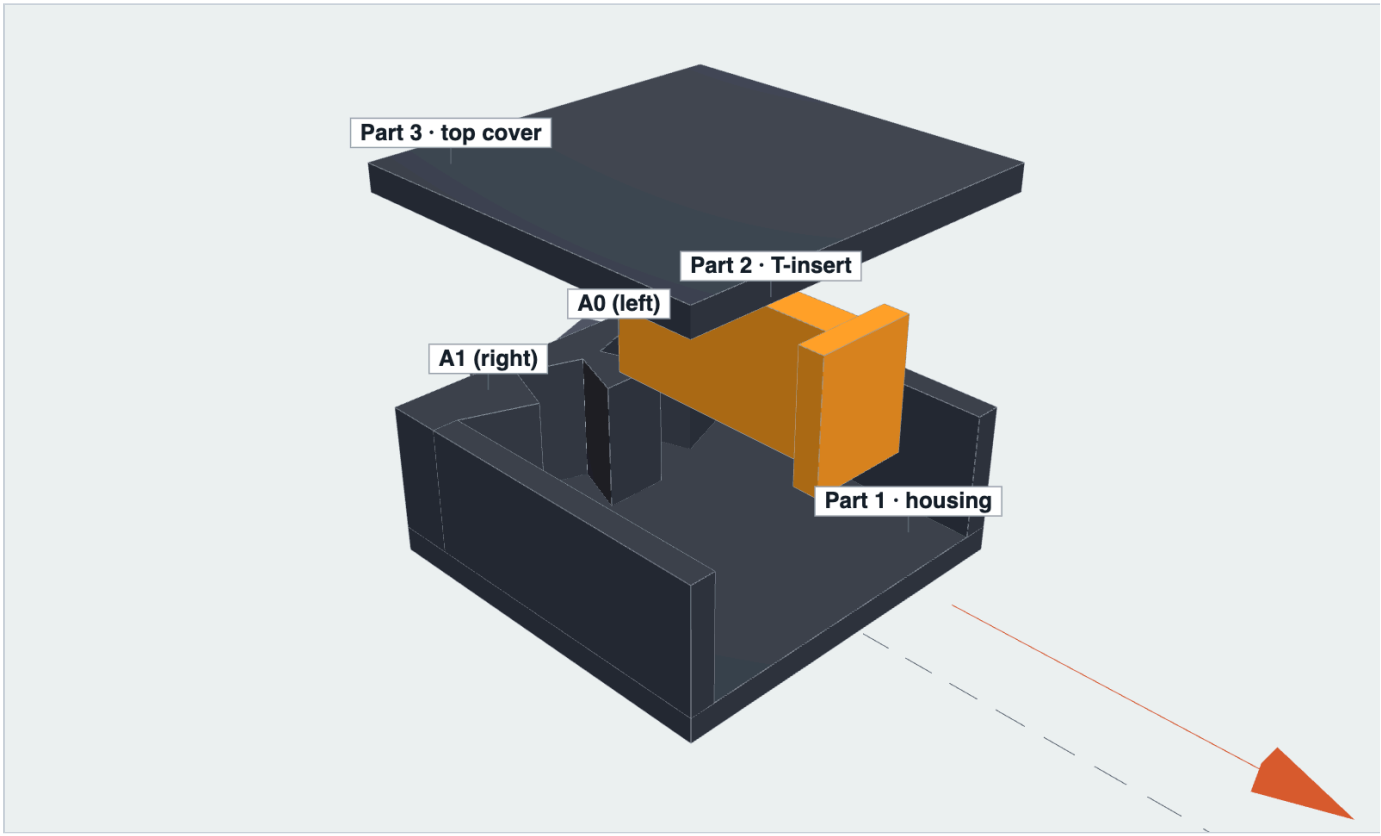
Shows the layout that sets the precision: sensors 10 mm apart and turned 20° outward, the 3 mm stem on the centre line, and the bar 20 mm in front of the lenses.

4. Side view



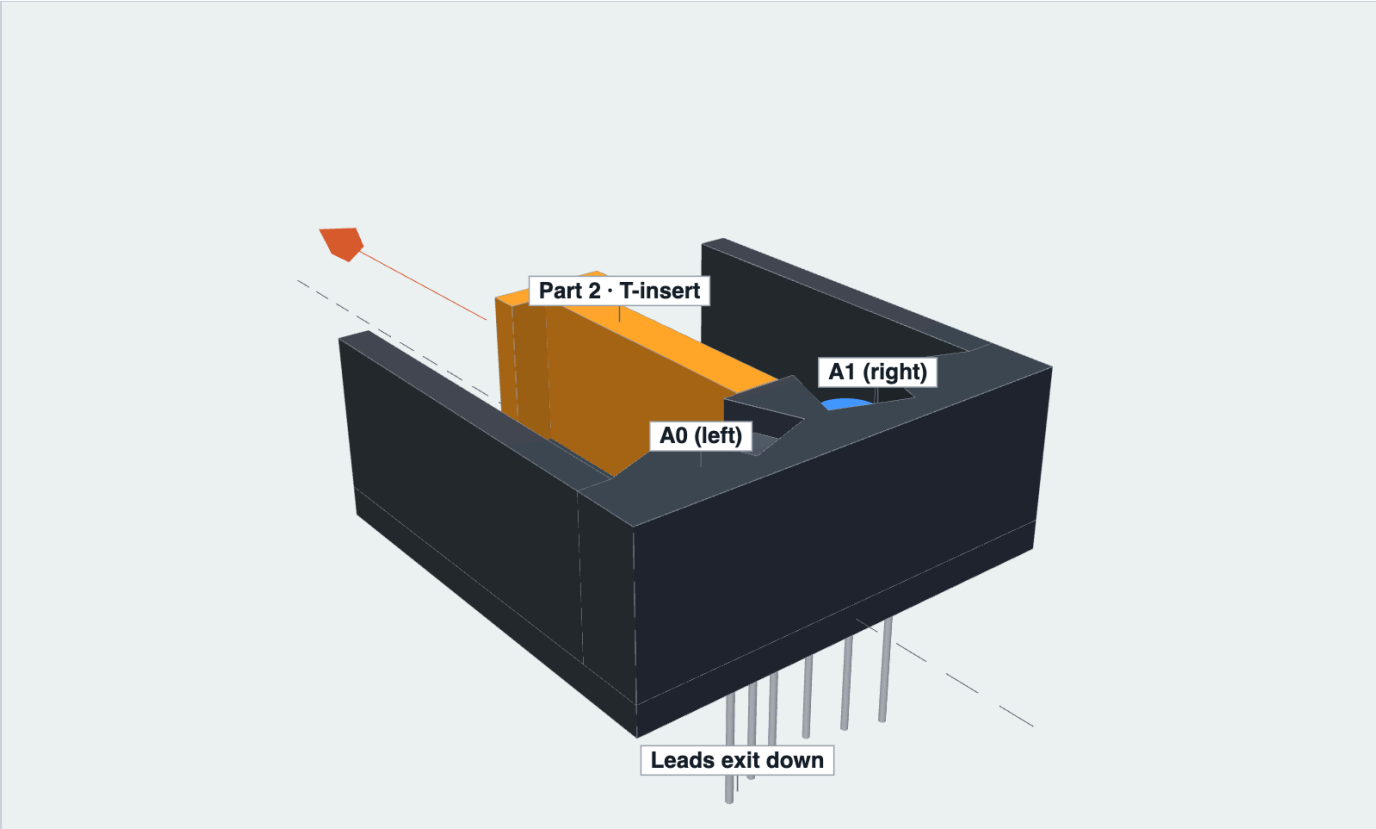
Radial depth: 29.5 mm overall, 25.5 mm inside the pipe wall. The sensors sit on the floor with their leads going down; the top cover closes the T-insert slot.

5. Exploded, 3/4 view



Assembly order: sensors into their pockets (leads through the floor), T-insert slides down into its 3.2 mm slot, top cover on with 2 × M2. Lift the cover to swap the bar (W 7 / 8 / 9).

6. Rear 3/4, cover removed



From behind and inside the satellite: the rear block with the two angled pockets, and the stem running forward between the lenses.